



VRIJE
UNIVERSITEIT
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PROGRAMMING ROBOTS WITH HIGHER-ORDER REACTIVE PROGRAMMING

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2025-2026

Promotor: Prof. Dr. Wolfgang De Meuter
Advisor: Bjarno Oeyen

Sciences and Bio-engineering Sciences



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Abstract

Reactive programming (RP) is a declarative programming paradigm based on data streams and the propagation of change. In a nutshell, if we have two variables x and y and another variable z that sums both x and y , it should be sufficient to update either x or y and z should be updated automatically.

There are several approaches to achieve reactive programming. One approach is to create a dedicated language. In this case the reactivity will be native to the language. Another option is to extend a language with reactive features, like

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Chapter 1

Introduction

Modern software is written in such a way that it is event-driven, or in other words, reactive. This means that applications reacts to an incoming event by creating subsequent events that arise as a reaction to the incoming event. Such an application is often called a reactive system. Reactive Systems are often implemented with callbacks or by implementing the Observer Pattern. However, this does not come without its drawbacks as this may lead to ‘inversion of control’ or ‘callback hell’. To alleviate this we introduce **Reactive Programming**. A programming paradigm that allows the user to write reactive systems in a declarative way. There are different ways to achieve Reactive Programming. One way is to create a reactive programming language, making the language itself reactive. Another is to extend a language with reactive features, either by introducing native reactivity or through a library. We will be looking at the former way and more specifically into the higher order reactive programming language **Haai**.

Additionally, we want to delve into the world of electronics. From maintaining your fridge’s temperature to controlling a drone’s altitude, electronics are deeply embedded in everyone’s day-to-day life. Writing software for these electronics is also a challenge, mainly due to the fact that using peripherals require using their associated libraries as well as using a low-level programming language such as C/C++ or Rust.

1.1 Contributions

1.2 Overview of this Dissertation

1.3 Approach

Chapter 2

State of the Art

2.1 Reactive Programming

2.2 RP languages targeting MCUs

2.3 Systems to program robots

2.4 Tierless Programming

2.5 Problem Statement

Chapter 3

μ -Remus

3.1 Haai & Remus

3.2 ROS & μ -ROS

3.3 Mapping RP to ROS & μ -ROS

3.4 μ -Remus: A C implementation

Chapter 4

Programming Robots with Reactors

4.1 Linking ROS and μ -ROS

Chapter 5

Tierless Haai

5.1 Haai to CHaai

Chapter 6

Evaluation

6.1 Evaluating Haai distribution

6.2 Evaluating Haai to CHaai

6.3 Evaluating CHaai on μ -Remus

Chapter 7

Conclusion & Future Work

